

Sunwoong Moon, Ph.D. Student

✉ anstjs7777@gm.gist.ac.kr

☎ +82-10-3371-7479



Research Interests

Goal : Human-synchronized and robust control of robots with optimality and stability.

- Phase variable based **synchronized** tracking control of exoskeleton.
- Development of real-time estimation and control models based on machine learning.
- Design of reinforcement learning-based agents for real-time estimation and control.
- Embedded system architecture for efficient data transmission and reception.
- Model-based optimal and robust control.

Education

- | | |
|----------------|--|
| 2024 – present | ■ Ph.D. Mechanical Engineering, GIST.
Dynamics and Biomechatronics LAB, (a.k.a. HurGroup). |
| 2022 – 2023 | ■ M.S. Mechanical Engineering, GIST.
Dynamics and Biomechatronics LAB, (a.k.a. HurGroup).
Thesis title: <i>Speed-Adaptive Gait Phase Estimation Using Dynamic Movement Primitives.</i> |
| 2016 – 2021 | ■ B.S. Mechanical Engineering, GIST.
Internship Dynamics and Biomechatronics LAB, (a.k.a. HurGroup).
Thesis title: <i>State-space Model Predictive Control for MIMO 2 DOF ball on plate system.</i> |
| 2013 – 2015 | ■ Sejong High School. |

Projects

Exoskeleton Control

- | | |
|------|---|
| 2025 | ■ Developed an intention-aware control strategy using IMU signals to detect the user's intention to initiate gait and provide timely assistive torque. |
| | ■ Attached 3 IMUs and a Multiflexor module to the exoskeleton (via I2C to MPU) and implemented an initial posture joint angle estimation model based on sensor fusion. |
| 2024 | ■ Designed and tested disturbance observer and gravity compensator. |
| | ■ Implemented functionality to receive and respond to motion commands sent from smart-watch/tablet via Bluetooth. |
| | ■ Collected joint angle trajectories for walking, sitting, and standing via motion capture and implemented interpolation-based tracking control for exoskeleton actuation. |
| 2023 | ■ Built a real-time distributed multi-core control framework using 1 MPU (4 cores) and 2 MCUs (2 cores) connected via UART. Implemented CAN-based motor control for 2 motors per MCU. |

Projects (continued)

Knee Assist Robot Control based on Free Energy Principle

- 2025  Developed a real-time control system for a knee exoskeleton combining Free Energy Principle (FEP)-based controller and PD controller integrated with a phase variable estimator on ROS2.
-  Implemented a rule-based finite state machine (FSM) for real-time classification of human locomotion phases, enabling reliable mode switching in wearable assistive devices.
- 2024  Designed a wearable knee-assist exoskeleton.
-  Implemented offline gait phase variable estimation algorithms using thigh angle from Motion Capture data in Julia (v1.9.3) to enhance phase-aware control of wearable robots.
- 2023  Constructed foundational code in Julia (v1.7.3) for simulation and analysis of FEP-based inference and control mechanisms.

Reinforcement Learning Term-project

- 2025  Designed and trained reinforcement learning agents (PPO, DQN) for controlling cart-pole, ball-on-beam, and double inverted pendulum systems. Derived system equations via the Euler-Lagrange formulation and used them to implement accurate, simulation-based learning environments.

KSPHM-KIMM Challenge

-  Participated in the KSPHM-KIMM Challenge, developing machine learning models (LSTM) to predict the remaining useful life (RUL) of bearings using temperature, vibration, and torque data.

Research Publications

Journal Articles

- 1 S. Moon and P. Hur, "State estimation of nonlinear system based on free energy principle," *Institute of Control, Robotics and Systems*, pp. 760–761, 2024.
- 2 W. Hong, N. A. Kumar, S. Moon, P. Hur, *et al.*, "A feasibility study of piecewise phase variable based on variable toe-off for the powered prosthesis control: A case study," *IEEE Robotics and Automation Letters*, vol. 8, no. 5, pp. 2590–2597, 2023.
- 3 S. Moon, W. Hong, and P. Hur, "A phase-shifting based human gait phase estimation for powered transfemoral prostheses," *대한기계학회 춘추학술대회*, pp. 81–81, 2022.
- 4 M. Cha, S. Moon, K. Cho, and P. Hur, "Dynamics and control methods for compass-gait walker: Tutorial of defining eom and simulation method for undergraduates," *Institute of Control, Robotics and Systems*, pp. 577–578, 2021.
- 5 S. Moon, M. Cha, K. Cho, and P. Hur, "Finding the optimal trajectory using a direct collocation method for an efficient and accurate compass-gait walker for undergraduates," *Institute of Control, Robotics and Systems*, pp. 579–580, 2021.

Conferences

- 2025  **ICROS, Poster**, LSTM-Based Real-Time Classification of Locomotion States Using Inertial Measurement Unit (IMU) Data.

Conferences (continued)

- **ICROS, Poster**, Phase Variable Estimation Based on Gait Pattern Adaptive Dynamic Movement Primitives.
- 2024 ■ **ICROS, Poster**, State Estimation of Nonlinear System Based on Free Energy Principle.
- 2023 ■ **대한생체역학회, Oral**, aDMP based Human Gait Pattern adaptive Phase Variable Estimation.
- 2022 ■ **대한생체역학회, Oral**, Piecewise Linear Labeling Method for Speed-Adaptability Enhancement in Human Gait Phase Estimation.
- **대한기계학회, Oral**, A Phase-Shifting Based Human Gait Phase Estimation for Powered Transfemoral Prostheses.
- 2021 ■ **ICROS, Poster**, Dynamics and control methods for compass-gait walker: Tutorial of defining eom and simulation method for undergraduates.
- **ICROS, Poster**, Finding the optimal trajectory using a direct collocation method for an efficient and accurate compass-gait walker for undergraduates.

Skills

Languages	■ Korean, English.
Softwares	■ Python, R, Julia, $\LaTeX$, C++, MATLAB, ROS2, AutoCAD.
Hardwares - Embedded	■ Raspberry Pi 4/5, Beagle Bone Black, Jetson Nano, Jetson Orin NX, Arduino, STM32, Teensy 4.1.
Hardwares - Sensors/Communications	■ MPU-9250 (9-axis IMU, I2C), Multiflexor (I2C), ADC converter (I2C), FSR (Analog Signal DAQ), AK series motors (T-MOTOR, CAN), MCU-to-MCU/MPU UART serial communication, Mobile Device-to-MCU Bluetooth serial communication.
Statistical Analysis	■ Repeated measures ANOVA, paired T-Test, R-square, RMSE.

Miscellaneous Experience

Awards and Achievements

- 2021 ■ **Best Thesis Award**, State-space Model Predictive Control for MIMO 2 DOF ball on plate system, GIST.
- 2022 ■ **Best Paper Award**, Piecewise Linear Labeling Method for Speed-Adaptability Enhancement in Human Gait Phase Estimation, 대한생체역학회.

References

Available on Request